Syllabus Fall 2026

MMG 2700, Foundations in R for Life Sciences, 3 Credit Hours (fall)

Meeting Time, Meeting Pattern, Location

Rowell 115

Instructor Name, Contact Information, Office Hours

Princess Rodriguez, PhD, princess.rodriguez@uvm.edu

Office Hours: Given Courtyard S453, by appointment only

Course Description

This course introduces students to programming in R with a focus on applications in microbial and cancer genomics. Through a combination of lectures, hands-on programming activities, and collaborative discussions, students will develop foundational computational skills for working with and interpreting biological datasets. Topics will include R syntax and programming fundamentals, data wrangling and manipulation with the tidyverse, exploratory data analysis and statistics, and data visualization using ggplot2. Students will also learn principles of reproducible research and effective communication of biological data through code, figures, and written interpretation. In the second half of the course, students will apply these skills to discipline-specific genomics applications, including microbiome composition analysis and the exploration of transcriptional and mutational profiles in cancer datasets. Students will critically evaluate primary research literature and use real-world biological datasets to practice computational approaches commonly used in modern genomics research. By the end of the course, students will be equipped with practical R programming and data analysis skills that support reproducible research in microbial genomics, cancer biology, and related areas of the life sciences.

Catamount Core/General Education (e.g., AH1; D1) or other requirements satisfied (if applicable). Please include corresponding outcomes information.

None

Course Learning Objectives/Outcomes

By the end of the course, students will be able to:

1. Operate R/RStudio independently to write, run, and document reproducible code.
2. Import, organize, and clean biological datasets in tabular formats.
3. Apply data wrangling and visualization techniques to summarize and interpret biological data.

4. Explore microbiology datasets to uncover microbiome diversity analysis.
5. Explore cancer biology applications such as gene expression, mutation profiling, and survival analysis.
6. Communicate findings effectively using R Markdown.

Technical support for students

Students, please read this technology checklist to make sure you are ready for classes.
https://www.uvm.edu/it/kb/student-technology-resources/

Students should contact the Helpline (802-656-2604) for support with technical issues.

Pre-requisites or co-requisites

BCOR 2300

Modality description/Outline (for Hybrid or Online courses)

All lectures will be delivered in person.

Required Course Materials:

There is no textbook required for this course. A laptop computer will be required to participate in this course.

Required platforms and software:

Each student is required to install R and RStudio on their personal computer. R is the underlying programming language used for statistical computing, while RStudio is an integrated development environment (IDE) that makes it easier to write and run R code. These tools are essential for the data analysis and visualization components of this course.

Brightspace, MS Teams, or other course sites:

For Brightspace information, students can access the following UVM Knowledge Base article:

https://www.uvm.edu/it/kb/article/brightspace-for-students/

Attendance Policy and Classroom Environment Expectations:

In this class, we will work together to develop a learning community that is inclusive and respectful. As a learning community we will seek to encourage and appreciate expressions of different ideas, opinions, and beliefs in the spirit of Our Common Ground. Meaningful and constructive dialogue is encouraged in this class. This requires mutual respect, willingness to listen, and open-mindedness to opposing points of view. Respect for individual differences and alternative viewpoints will be maintained at all times in this class. Conduct that substantially or repeatedly disrupts the ability of faculty and instructors to teach and the ability of students to engage may result in my asking a student to temporarily leave the classroom. See Undergraduate Catalogue - Classroom Code of Conduct.

Lecture Attendance Policy

Attendance in lecture is expected. Class participation using your laptop with internet-connection will be a component of the course grade and used in most lectures to encourage participation and interaction.

Grading Criteria/Policies

The grade will be determined as follows:

Project 1	15%
Project 2	25%
Quizzes	25%
Team-Based Discussion Questions	25%
Participation	10%
TOTAL	**100%**

A+ | 97 - 100	A | 94 - 96	A- | 90 - 93	
B+ | 87 - 89	B | 84 - 86	B- | 80 - 83	
C+ | 77 - 79	C | 74 - 76	C- | 70 - 73	
D+ | 67 - 69	D | 64 - 66	D- | 60 - 63	F | 59 or below

Academic Integrity and AI:

This course assumes that all work submitted by students will be generated by the students themselves, working individually or in groups. Students should not have another person/entity do the writing of any substantive portion of an assignment for them, which includes hiring a person or a company to write assignments and using artificial intelligence tools like ChatGPT.

Assessments (Graded Work):

Below are descriptions of the assessments that will be used to measure the students understanding of course concepts.

Team-Based Discussion Questions (25%):

Critical evaluation of scientific literature and effective scientific communication are essential skills for researchers working with biological data. Throughout the semester, students will participate in team-based learning (TBL) activities centered on assigned primary research articles relevant to genomics and bioinformatics. Prior to each TBL session, students will be expected to read the assigned article and prepare to discuss the study with their peers. During class, students will work collaboratively in teams to answer discussion questions based on the assigned paper. Questions may require students to evaluate the experimental design, interpret figures and results, assess computational approaches, and connect the findings to concepts introduced throughout the course. These activities are designed to strengthen students' ability to critically read scientific literature, interpret biological and computational results, communicate their reasoning, and engage in collaborative scientific discussion.

Project 1 (15%):

Students will complete an individual differential expression analysis project using R and RStudio. Students will apply the programming and data analysis skills developed throughout the course to analyze a dataset, including data exploration, visualization, and differential expression analysis. Students will submit their work as a reproducible R Markdown document that clearly presents their code, results, figures, and interpretation of findings. The project will assess students' ability to organize and analyze biological data, generate appropriate visualizations, interpret differential expression results, and communicate their analysis clearly and reproducibly. A detailed grading rubric will be provided to outline expectations for code organization, reproducibility, analysis, visualization, interpretation, and overall presentation.

Project 2 (25%):

As a capstone to the course, students will complete a team-based project that applies R programming skills to real-world questions in either microbial or cancer genomics. Working in pairs, students will design and carry out an analysis that demonstrates their ability to wrangle, visualize, and interpret biological data. Projects will involve key computational steps such as quality control, exploratory and differential analysis, and the creation of clear, publication-style plots in R/RStudio. Final results will be presented to the class in an oral presentation format using Microsoft PowerPoint. Each presentation will be evaluated on several dimensions: clarity and depth of the background overview, soundness of experimental design and quality control procedures, interpretation of results and findings, effective use of plots and figures, and overall delivery. Attention to grammar, formatting, and professional presentation style will also be assessed, along with evidence of balanced contributions from both partners. A detailed grading rubric will be provided to outline expectations for code organization, reproducibility, analysis, visualization, interpretation, and overall presentation.

Class Participation (10%):

Most class sessions will include a hands-on component that requires students to bring their laptops to class, follow along with demonstrations, complete programming activities, and answer questions. Active participation in these activities is expected and will provide students with opportunities to practice R programming skills and receive guidance in real time. At the end of most class sessions, students will be required to knit and submit their completed R Markdown file for participation points. These submissions will serve as evidence of participation and completion of the in-class activities.

Quizzes (25%)

There will be at least five quizzes assigned throughout the semester. Each quiz will be made available at least 24-hours before the due date and time specified. The quiz may include multiple choice or short answer questions. Late quizzes will not be accepted. A quiz is considered late if it is submitted after the date and time specified.

Additional Policies

Lived Name and Pronoun Information

The UVM Directory includes fields for indicating your lived name and your pronouns. Lived names (preferred names, names in use) are names that an individual wants to be known by in the University community. Entering your pronouns is strongly encouraged to help create a more inclusive and respectful campus community. To update your information, login to the UVM Directory. A preview box will allow you to see how this information will appear in other systems used on campus such as Microsoft Teams and Blackboard.

More information about how to make changes to your lived name and pronouns is available in the Knowledge Base.

AI Use Policy

Since writing, analytical, and critical thinking skills are part of the learning outcomes of this course, all writing assignments should be prepared by the student. Developing strong competencies in this area will prepare you for a competitive workplace. Therefore, AI-generated submissions are not permitted and will be treated as plagiarism.

Research and Citation Help

For help selecting research topics, finding information, citing sources, and more, ask a librarian. The UVM Libraries are eager to help. You may ask questions by phone, e-mail, chat, or text, or make an appointment for an individual consultation with a librarian.

Howe Library: https://library.uvm.edu/askhowe
Dana Medical Library: https://dana.uvm.edu/help/ask
Silver Special Collections Library: https://specialcollections.uvm.edu/help/ask

Course Evaluation

All students are expected to complete an evaluation of the course at its conclusion. These evaluations are anonymous and confidential, and the information gained, including constructive criticisms, will be used to improve the course.

General statement regarding potential changes during the semester:

http://catalogue.uvm.edu/
The University of Vermont reserves the right to make changes in the course offerings, mode of delivery, degree requirements, charges, regulations, and procedures contained herein as educational, financial, and health, safety, and welfare considerations require, or as necessary to be compliant with governmental, accreditation, or public health directives.

Intellectual Property Statement/Prohibition on Sharing Academic Materials

Students are prohibited from publicly sharing or selling academic materials that they did not author (for example: class syllabus, outlines or class presentations authored by the professor, practice questions, text from the textbook or other copyrighted class materials, etc.); and students are prohibited from sharing assessments (for example homework or a take-home examination). Violations will be handled under UVM's Intellectual Property policy and Code of Academic Integrity.

Tips for Success

Course-specific study/preparation tips
Here are a few resources for students on remote/online learning:

- Checklist for success in https://learn.uvm.edu/about/support-for-students/checklist-online-credit-courses/
- Academic support for online courses: https://www.uvm.edu/academicsuccess/online-learning-student-resources-remote-instruction

Helpful resources other than the professor (e.g., Undergraduate/Graduate Writing Center, Supplemental Instruction, Learning Co-op tutors, supplemental course materials)

Student Learning Accommodations:

In keeping with University policy, any student with a documented disability interested in utilizing ADA accommodations should contact Student Accessibility Services (SAS), the office of Disability Services on campus for students. SAS works with students and faculty in an interactive process to explore reasonable and appropriate accommodations, which are communicated to faculty in an accommodation letter. All students are strongly recommended to discuss with their faculty the accommodations they plan to use in each course. Faculty who receive Letters of Accommodation with Disability Related Flexible accommodations that go beyond the default accommodations will need to fill out the Disability Related Flexibility Agreement. Any questions from faculty or students on the agreement should be directed to the SAS specialist who is indicated on the letter.

Contact SAS:
A170 Living/Learning Center;
802-656-7753
access@uvm.edu
www.uvm.edu/access

Important UVM Policies

(You may provide these in a separate document posted in Brightspace if you wish)

Academic Integrity:

The Academic Integrity policy addresses plagiarism, fabrication, collusion, and cheating.

Code of Student Conduct:

UVM's Code of Student Conduct outlines conduct expectations as well as students' rights and responsibilities.

FERPA Rights Disclosure:

The purpose of UVM's FERPA Rights Disclosure is to communicate the rights of students regarding access to, and privacy of their student educational records as provided for in the Family Educational Rights and Privacy Act (FERPA) of 1974.

Final Exam Policy:

The University final exam policy outlines expectations during final exams and explains timing and process of examination period.

Grade Appeals:

If you would like to contest a grade, please follow the procedures outlined in this policy.

Grading:

This link offers information on grading and GPA calculation.

Religious Holidays:

Religions may be practiced in many different ways, and can impact participation in classes variably. Students have the right to practice the religion of their choice. Each semester students should submit in writing to their instructors as early as possible and at least one week prior to their documented religious holiday the date(s) of the conflict or absence. Faculty must permit students who miss work or exams for the purpose of religious observance to make up this work. The complete policy is here.

Promoting Health & Safety:

The University of Vermont's number one priority is to support a healthy and safe community:

Center for Health and Wellbeing offers a variety of resources, including resources to support physical and mental health and food security, including Counseling and Psychiatry Services (CAPS). The CAPS direct phone line is (802) 656-3340.

C.A.R.E. If you are concerned about a UVM community member or are concerned about a specific event, we encourage you to contact the Dean of Students Office (802-656-3380). If you would like to remain anonymous, you can report your concerns online by visiting the C.A.R.E. Team website.